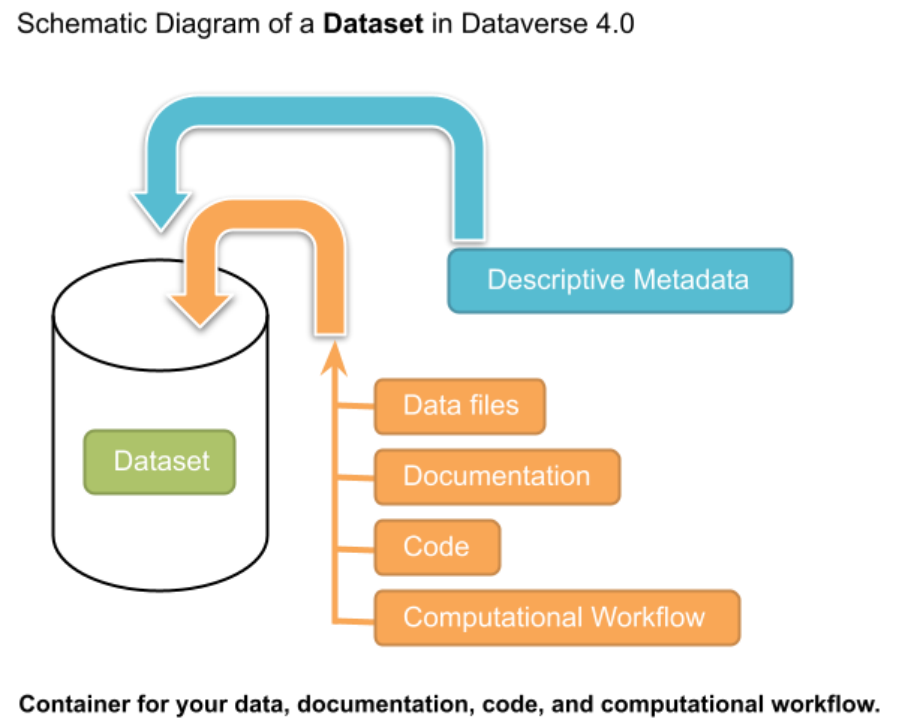


SDM Crash Course #5: Data Management Practices



What are we doing today?



Code and Data Management:

- What it is and why it is important
- Common deficiencies in ecological modeling
- Best practices
 - Data
 - Code
 - Taxonomy
 - Reporting

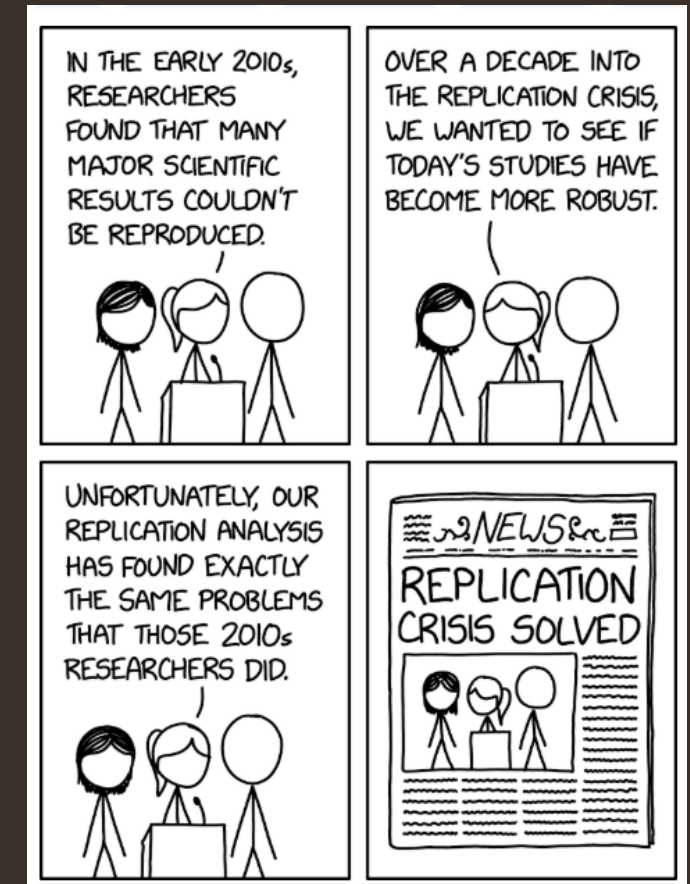
Data and Code Management

Data management: “The development, execution, and supervision of plans, policies, programs, and practices that deliver, control, protect, and enhance data and information throughout their lifecycles” (Data Administration Management Association)

Code management: The organization, presentation, and maintenance of code that makes up a given product (here, scientific research)

Reproducibility: The ability to independently support the claims/findings of a given study (requires change/will not be the exact same as the original)

Replicability: The ability to **exactly** recreate the outputs of a given study



My Soapbox



Data and Code Management

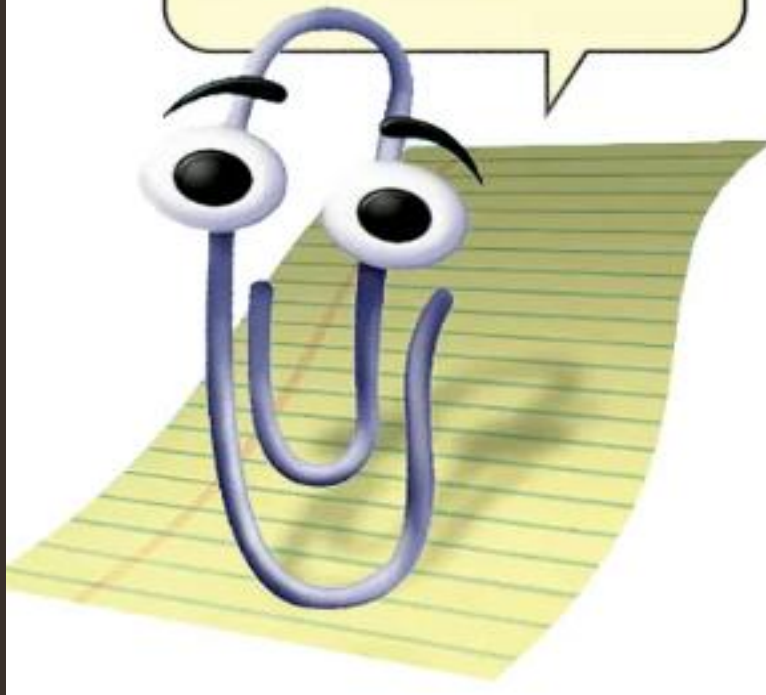
In my opinion, data and code management are a matter of research ethics (**esp. for SDMs**)

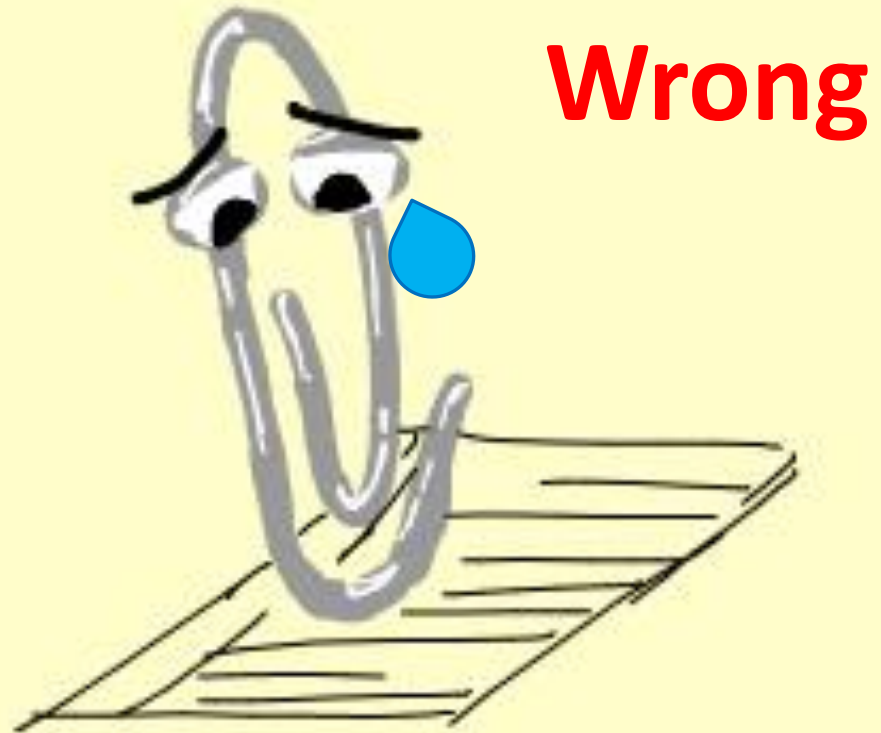
- Crux of the work is in the data and code used
- Most SDM work is *intended* to provide insights/information for practical applications
 - Decision making, interventions, conservation/mitigation management
- Presenting our work in interpretable, replicable ways makes our work clear and transparent
- Failing to do so weakens our science, which could ultimately misguide management practices

Thus, as ecologists (or just lovers of our natural world), it is our duty to adhere to good code and data management practices



So, this means all
current works manage
and present data and
code in interpretable,
accessible ways, right?





A Real Example

Shameless plug of my own article on SDMs applied to snail hosts of parasites

Openly accessible:

- Response data (21%)
- Predictor data (10%)
- Code (6%)

The above doesn't even take into account the **quality** of accessible data/code (does it have metadata, annotations, is it machine readable, etc.)

No studies explicitly reference a global taxonomy, only 17% implicitly use one





How Can I be a Good Practitioner?

Data Practices: Data Storage

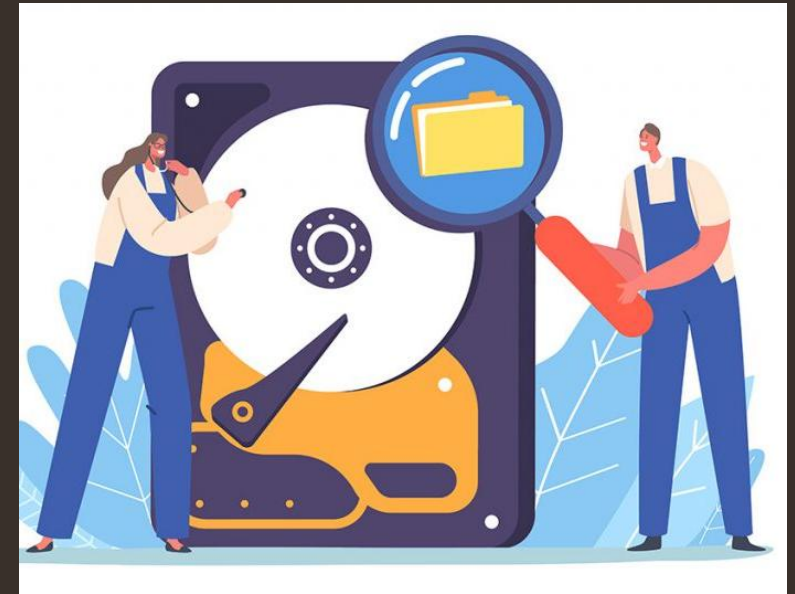
Storage in online settings, not just on hardware

- Prevents issues relating to mechanical failure or damage
- Easier to share: collaborators, peers, lab group, etc.

Allows for easier updating of files than manual backups

- Do periodically with the click of a button, or setup machine to automatically backup

Ensures preservation far after you are done with a project/have new hardware



Data Practices: Metadata

Metadata: “Data which provides information about other data” (Merriam-Webster.com Dictionary, 2026)

- Ex: What do columns mean? What type of data is stored in each column/file? What is the file naming convention? What is each file used for? Etc.

International Organization for Standardization (ISO) compliance

- Data is machine readable (i.e. standard format), traceable (track of uploads and edits), and with some form of protection/security from unauthorized edits

Standardized metadata allows people to read your data and storage system easily

Example formats (Field): Ecological Metadata (Ecology [general]), Darwin Core (Biodiversity), ISO 19115 (Geographic), Climate and Forecast Conventions (Earth Science)

Contact Information

Rough description of
how data was
gathered

Information on what
each file contains

Where these
files/data can be
accessed

```
</publisher>
  <individualName>
    <givenName>Nicholas</givenName>
    <givenName>A</givenName>
    <surName>Galle</surName>
  </individualName>
  <organizationName>University of Notre Dame</organizationName>
  <positionName>PhD Student</positionName>
  <electronicMailAddress>nickgalle99@gmail.com</electronicMailAddress>
  <userId directory="https://orcid.org">0000-0002-9833-453X</userId>
</publisher>
</methods>
<methodStep>
  <description>
    <para>Studies screened to make the datasets herein were collected through a systematic literature review to identify studies that
    species modeled, SDM methodologies applied, and adherence to Open Access and best modeling practices. </para>
  </description>
</methodStep>
</methods>
<project>
  <title>Synthesis and analysis of species distribution models applied to snail intermediate hosts.</title>
  <personnel>
    <individualName>
      <givenName>Nicholas</givenName>
      <givenName>A</givenName>
      <surName>Galle</surName>
    </individualName>
    <organizationName>University of Notre Dame</organizationName>
    <positionName>PhD Student</positionName>
    <electronicMailAddress>nickgalle99@gmail.com</electronicMailAddress>
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    <role>First Author</role>
  </personnel>
</project>
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  <entityDescription>Data table from supplementary file 1, containing extracted metadata from studies applying SDMs to snail intermediate
  </entityDescription>
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        <quoteCharacter>"</quoteCharacter>
      </simpleDelimited>
    </textFormat>
  </dataFormat>
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    <online>
      <url function="download">https://ezeml.edirepository.org/user-data/Nicholas_Galle-8a5d935ed6a998117c2bcbeb51056a34/uploads/Supp
    </online>
  </distribution>
</dataTable>
```

Name and full description of what is included in this column

What column entries mean

```
▼<attribute id="ee4095a3-0201-464f-9345-e0093138ce17">
  <attributeName>Schisto</attributeName>
  <attributeLabel>Schisto</attributeLabel>
  <attributeDefinition>This is the entry designating whether the studies model species that transmit schistosomiasis.</a
  <storageType typeSystem="XML Schema Datatypes">string</storageType>
  ▼<measurementScale>
    ▼<nominal>
      ▼<nonNumericDomain>
        ▼<enumeratedDomain enforced="yes">
          ▼<codeDefinition>
            <code>M</code>
            <definition>Indicates that some, but not all, of the species modeled in this study transmit schistosomiasis.
          </codeDefinition>
          ▼<codeDefinition>
            <code>N</code>
            <definition>Indicates that none of the species modeled in this study transmit schistosomiasis.</definition>
          </codeDefinition>
          ▼<codeDefinition>
            <code>Y</code>
            <definition>Indicates that all of the species modeled in this study transmit schistosomiasis.</definition>
          </codeDefinition>
        </enumeratedDomain>
      </nonNumericDomain>
    </nominal>
  </measurementScale>
</attribute>
```

Metadata from Galle et al., 2026

Data Practices: Metadata Example

Example: A PhD student graduates and leaves data and code for a work-in-progress project on their lab computer. A future student inherits the project, but does not understand how the data was stored (column name and entries are non-informative/not uniform), the file naming convention (which file does what, and where does it go in the modeling process), or what code is even relevant to the project (files spread across different directories)

- Cost: Many hours of work, and the potential that curated data is lost or misused



Data Practices: Open Access

Open Access: “A broad international movement that seeks to grant free and open online access to academic information, such as publications and data.” (openaccess.nl)

Making our publications, data (response, predictor, outputs), and code openly accessible makes our science more visible and transparent

- People know what inputs you used to get your outputs (they don’t just have to trust you)
- People can access and, with proper credit, use your data (meta-analyses, etc.)
- Some funding requires compliance to Findable, Accessible, Interoperable, Reusable (FAIR) principles

Some cases where full open access is not feasible

- Patient data, sensitive information, etc.





4.	Embargo on data due to commercial restrictions	The data that support the findings will be available in [repository name] at [URL / DOI link] following an embargo from the date of publication to allow for commercialization of research findings.
5.	Data available on request due to privacy/ethical restrictions	The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.
Data openly available in a public repository that issues datasets with DOIs		The data that support the findings of this study are openly available in [repository name e.g "figshare"] at http://doi.org/[doi], reference number [reference number].
Data openly available in a public repository that does not issue DOIs		The data that support the findings of this study are openly available in [repository name] at [URL], reference number [reference number].
Data derived from public domain resources		The data that support the findings of this study are available in [repository name] at [URL/DOI], reference number [reference number]. These data were derived from the following resources available in the public domain: [list resources and URLs]

Data availability

Data will be made available on request.

Often, really isn't

Outputs supporting the exploration of zero-inflated integrated species distribution models

May 7, 2026

[View Data Release](#)

No inputs or code, just outputs



Coding Practices

Our code practices don't really matter, as long as the underlying approach is rigorous, right?



Script Name: glm_me_fun (1) (1).R

```
51 ▾ auc<-function(pred,obs){pred<-as.numeric(pred)
52   obj<-pROC::roc(obs,pred)
53 ▲   return(as.numeric(pROC::auc(obj)))}
54   b_pred<-predict(maxnet_model, pb_data,type="cloglog")
55   b_auc<-auc(b_pred, pb)
56   imp<-numeric(ncol(pb_data))
57   names(imp)<-colnames(pb_data)
58 ▾ for(v in colnames(pb_data)){data<-pb_data
59   data[[v]]<-sample(data[[v]])
60   pred<-predict(maxnet_model,data,type="cloglog")
61   auc<-auc(pred,pb)
62 ▲   imp[v]<-baseline_auc-auc}
63   imp_pct<-100*imp/sum(imp)
64   sort(imp_pct,decreasing=TRUE)
```

Descriptive script
name

Script Name: SDM_evaluation_functions.R

Descriptive object
names

Spacing between
text

Comments on what
the code is doing

```
146 # Define your function to compute AUC
147 get_auc <- function(pred, obs) {
148     pred <- as.numeric(pred)
149     roc_obj <- pROC::roc(obs, pred)
150     return(as.numeric(pROC::auc(roc_obj)))
151 }
152
153 # Baseline predictions based on training data
154 baseline_pred <- predict(maxnet_model, pb_data, type = "cloglog")
155 baseline_auc <- get_auc(baseline_pred, pb)
156
157 # Initialize importance vector
158 var_imp <- numeric(ncol(pb_data))
159 names(var_imp) <- colnames(pb_data)
160
161 # Permutation importance loop
162 for (v in colnames(pb_data)) {
163     perm_data <- pb_data
164     # Shuffle that variable
165     perm_data[[v]] <- sample(perm_data[[v]])
166     # Predict with permuted data
167     perm_pred <- predict(maxnet_model, perm_data, type = "cloglog")
168     perm_auc <- get_auc(perm_pred, pb)
169     # Importance = drop in AUC
170     var_imp[v] <- baseline_auc - perm_auc
171 }
172 # Normalize (optional, to sum to 100)
173 var_imp_pct <- 100 * var_imp / sum(var_imp)
174 # View results
175 sort(var_imp_pct, decreasing = TRUE)
```


Coding Practices

Annotate R script and follow readability principles (spacing, object and file naming, etc.)

- Allows people to replicate or build upon your work with ease

Version control (R ***and*** packages)

- Check out 'renv', 'usethis', and 'gert' R packages
- Set seed, if using stochastic modeling approach

Save, load, and present critical components of R script independently

- Data, model object, etc.

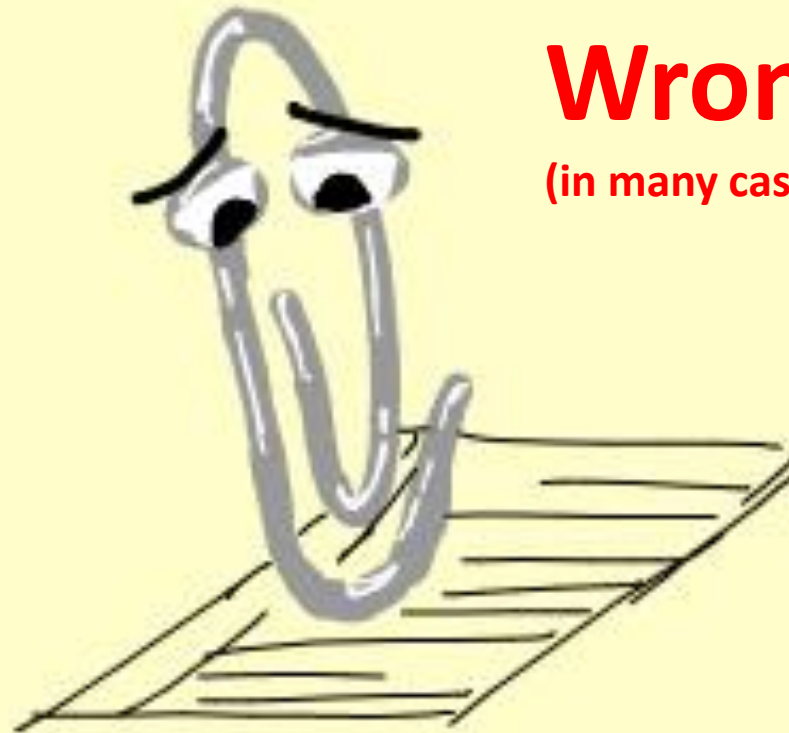
Keep track of, and detail, what each script does in metadata

Present code for figure making as well



Taxonomy Practices

We can just give a species name, and maybe get fancy with it and add the describer/date, right?



Wrong
(in many cases)



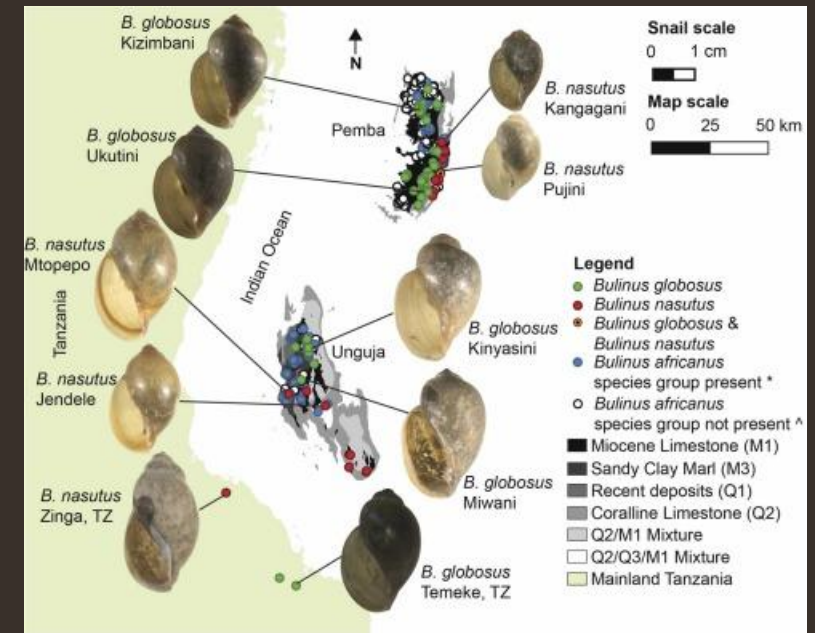
Taxonomy Practices

Taxonomic distinctions are made based on a variety of factors

- Gross morphology (size, coloration, beak shape, etc.)
- Microscopic morphology
- Genetic differences
- Other (sometimes based on the above PLUS geography)

As approaches become more refined, taxonomic reshuffling often occurs

This can result in ambiguity regarding the actual taxa studied, even in recent history for “troublesome” taxa (e.g., freshwater mollusks)



Taxonomy Practices

WoRMS taxon details

★ ***Bulinus truncatus* (Audouin, 1827)**

AphiaID 716338 (WoRMS species.org:taxname:716338)

Classification Biota > ★ Animalia (Kingdom) > ★ Mollusca (Phylum) > ★ Gastropoda (Class) > ★ Euthyneura (Infraclass) > ★ Tectipleura (Subterclass) > ★ Hygrophila (Superorder) > ★ Lymnaeoidea (Superfamily) > ★ Bulinidae (Family) > ★ *Bulinus truncatus* (Species)

Status accepted

Rank Species

Parent ★ *Bulinus* O. F. Müller, 1781

Original name ★ *Physa truncata* Audouin, 1827

Synonymised names

- ★ *Bulinus* (*Bulinus*) *truncatus* (Audouin, 1827) · unaccepted
- ★ *Bulinus* (*Isidora*) *dybowskii* P. Fischer, 1891 · alternative representation
- ★ *Bulinus* (*Isidora*) *truncatus* (Audouin, 1827) · alternative representation
- ★ *Bulinus* (*Isidora*) *truncatus rohlfsi* (Clessin, 1886) · unaccepted (junior synonym)
- ★ *Bulinus* (*Isidora*) *truncatus truncatus* (Audouin, 1827) · alternative representation
- ★ *Bulinus dybowskii* P. Fischer, 1891 · unaccepted
- ★ *Bulinus guernei* (Dautzenberg, 1890) · unaccepted (junior synonym)
- ★ *Bulinus hemprichii* (Ehrenberg, 1831) · unaccepted (junior synonym)
- ★ *Bulinus rohlfsi* (Clessin, 1886) · unaccepted (a junior synonym)
- ★ *Bulinus sericinus* (Jickeli, 1874) · unaccepted (a junior synonym)
- ★ *Bullinus* (*Isidora*) *mussolinii* Piersanti, 1940 · unaccepted (a junior synonym)
- ★ *Isidora broccchii* Ehrenberg, 1831 · unaccepted > junior subjective synonym
- ★ *Isidora guernei* Dautzenberg, 1890 · unaccepted (junior synonym)
- ★ *Isidora hemprichii* Ehrenberg, 1831 · unaccepted
- ★ *Isidora sericina* Jickeli, 1874 · unaccepted (a junior synonym)
- ★ *Physa* (*Isidora*) *broccchii* (Ehrenberg, 1831) · unaccepted > superseded combination
- ★ *Physa* (*Isidora*) *broccchii* var. *approximans* Mousson, 1876 · unaccepted > junior subjective synonym
- ★ *Physa* (*Isidora*) *lirata* Mousson, 1874 · unaccepted > superseded combination (original name; junior homonym of...)
- ★ *Physa* (*Isidora*) *vaneyi* Germain, 1907 · unaccepted
- ★ *Physa maresi* Bourguignat, 1862 † · unaccepted > junior subjective synonym
- ★ *Physa mutandaensis* Preston, 1913 · unaccepted (junior synonym)
- ★ *Physa rohlfsi* Clessin, 1886 · unaccepted (a junior synonym)
- ★ *Physa truncata* Audouin, 1827 · unaccepted (original combination)

Current accepted

Unaccepted now, but used at some point

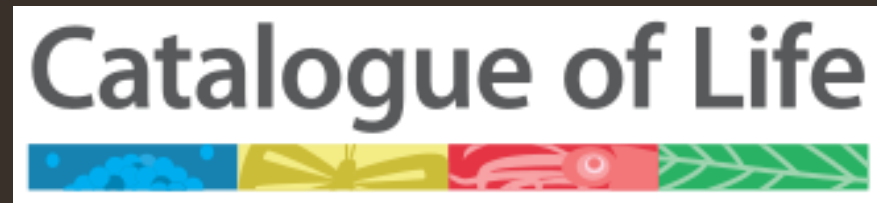
Taxonomy Practices

Solution: Use a globally accepted taxonomic system

These provide ID numbers for each distinct taxonomy, which stay with that distinction regardless of whether it is the currently accepted term

This allows future researchers to **follow what species was actually modeled in a given study**, regardless of any future taxonomic changes

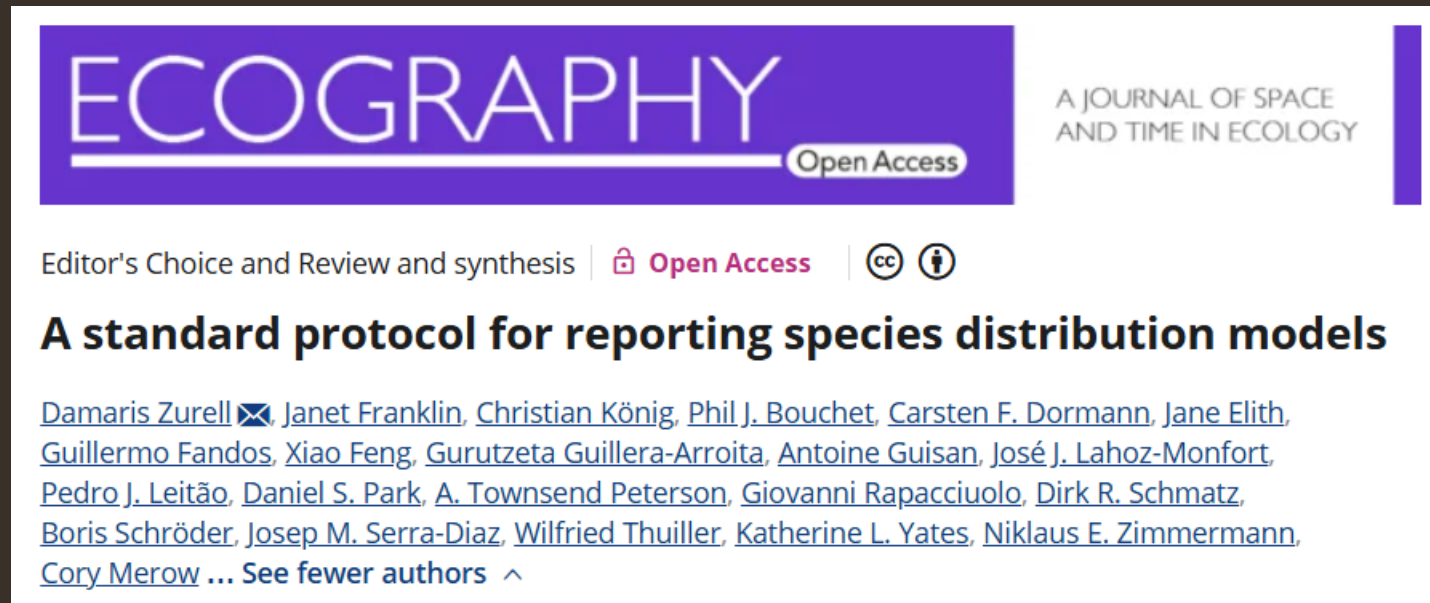
Examples (Taxa): World Registry of Marine Species (marine and freshwater), Integrated Taxonomic Information System (All, focused on North America), Global Biodiversity Information Facility Taxonomic Backbone (All), Catalogue of Life (All), Molluscabase (Mollusca)



Reporting Practices

Zurrell et al., 2020 present a standard protocol for reporting SDMs (ODMAP)

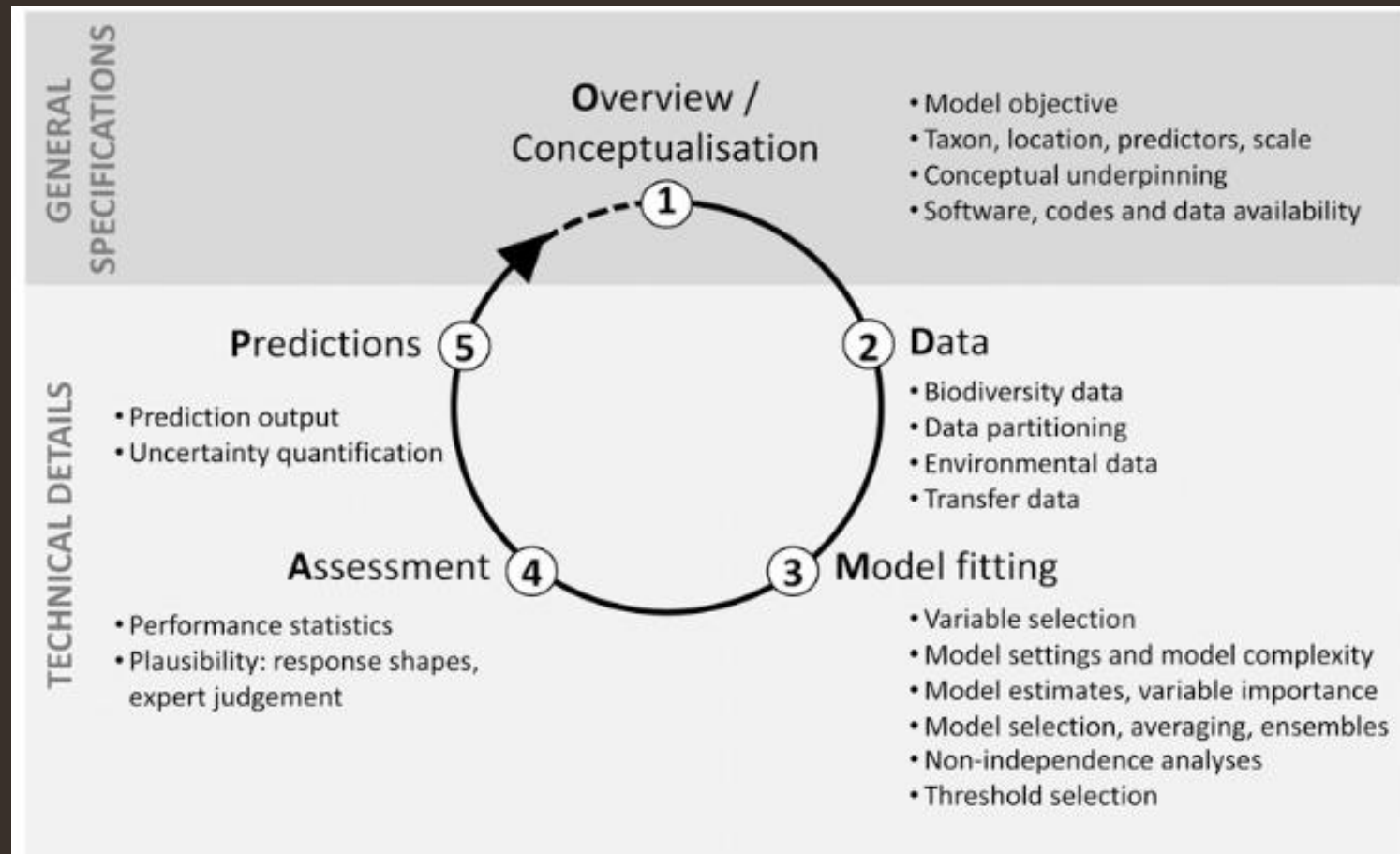
- Overview
- Data
- Model
- Assessment
- Prediction



See also



Reporting Practices (Elaborated)



Bouchet & Meeuwig (2015). Drifting baited stereo-videography: A novel sampling tool for surveying pelagic wildlife in offshore marine reserves. *Ecosphere*, 6: art137.

ODMAP element	Contents
OVERVIEW	
<i>Authorship</i>	• Authors: Phil J. Bouchet, Jessica J. Meeuwig • Contact email: pjbouchet@gmail.com • Title: Drifting baited stereo-videography: A novel sampling tool for surveying pelagic wildlife in offshore marine reserves • DOI: 10.1890/ES14-00380.1
<i>Model objective</i>	• Objective: Mapping/interpolation. • Target outputs: Maps of relative probability of presence
<i>Taxon</i>	Pelagic fishes, sharks, and other marine vertebrates
<i>Location</i>	Perth Canyon, Western Australia, Australia.
<i>Scale of analysis</i>	• Spatial extent (Lon/Lat): Longitude 115° E – 115.25° E, Latitude 31.90° S – 32.04° S • Spatial resolution: 350 m • Temporal extent/time period: Autumn (March to May) 2013 • Type of extent boundary: Rectangular
<i>Biodiversity data overview</i>	• Observation type: Field survey • Response/Data type: Presence-only
<i>Type of predictors</i>	Topographic, climatic
<i>Conceptual model / Hypotheses</i>	• Hypotheses about species-environment relationships: There is increasing evidence that complex seabed features influence the distribution of marine vertebrates. We used measures of seabed depth, curvature, slope, rugosity, and topographic complexity as predictor variables for fishes and sharks. We also tested sea surface temperature (and its variance).
<i>Assumptions</i>	We assumed that: • Relevant ecological drivers (or proxies) of species distributions are included. • Detectability does not change across transects or habitat gradients. • Species are at (pseudo-) equilibrium with their environment. • Species show largely similar responses to the bait and the sampling instrument. Sampling is adequate and representative (and any biases are accounted for/corrected).
<i>SDM algorithms</i>	• Algorithms: We fitted MaxEnt and single-visit site-occupancy models to the field data. MaxEnt was chosen due to competitive performance on small

MODEL	
<i>Variable pre-selection</i>	The choice of initial covariates was made as a compromise between their availability and their ecological relevance as indirect proxy of species distributions. Only weakly correlated covariates were included in each model.
<i>Multicollinearity</i>	Multicollinearity between predictors was investigated using variance inflation factors and Spearman rank correlation coefficients. When variables were strongly related ($ rs > 0.5$ and/or $VIF > 2$), we only retained one from each pair to minimize the possibility of over-fitting.
<i>Model settings</i>	• Model settings: Default settings for MaxEnt (but tuned regularisation parameter) and single-visit site-occupancy models (linear terms). • Model settings for extrapolation: Predictions bounded by clamping in MaxEnt.
<i>Model estimates</i>	Covariate importance was calculated as the sum of model weights for models containing the covariate. Model-averaged parameter estimates from occupancy models are shown in Table 2.
<i>Model averaging / Ensembles</i>	We took an information-theoretic approach to model averaging for each model class, considering all combinations of covariates. Within model classes, candidate models were ranked based on their AICc scores, with models within three units of lowest-AICc model retained and averaged according to their corresponding Akaike weights. Consensus predictions were then obtained across model classes using weighted ensemble means.
<i>Non-independence</i>	Spatial autocorrelation was assessed using spline correlograms in the R package “ncl” (Bjornstad, 2016).
ASSESSMENT	
<i>Performance statistics</i>	• Performance statistics estimated on training data: Model performance was assessed based on the average true skill statistic (TSS) from 50 bootstrap runs, calculated based on a TSS-maximisation threshold.
<i>Plausibility checks</i>	• Response plots: We used partial dependence plots to check the ecological plausibility of fitted relationships in MaxEnt models.

PREDICTION	
<i>Prediction output</i>	Predictions of relative probability of presence expressed on a continuous scale.
<i>Uncertainty quantification</i>	We compared model outputs to assess algorithmic uncertainty, and applied an ensemble approach averaging over two different model algorithms.

To Recap

Make sure your work that involves data analysis and coding is accessible and replicable

- See FAIR/ISO and similar frameworks for further guidance
- Data and code is openly accessible (when appropriate) and machine readable
- Properly annotated code (readability, comments) and data (comprehensive metadata)

Use an accepted global taxonomy, particularly for some taxa

Make clear and report all model objectives, data processing, and outcomes

- See ODMAP and similar frameworks for guidance

To Recap

Police yourself and others when it comes to data/code/reporting practices!

- Science largely relies on self- and peer-policing to ensure findings are rigorous
- Modeling studies that do not clearly present data and code **are not transparent or replicable**
- If we don't hold ourselves and each other accountable on this, we are undermining our own work

